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Academic year 2023-2024 (Even Sem)

DEPARTMENT OF

ELECTRONICS AND INSTRUMENTATION ENGINEERING

Date	18th June 2024	Maximum Marks	50
Course Code	EI243AI	Duration	90 Min
Sem	IV Semester	**Test-1**	
MICROCONTOLLER & PROGRAMMING (Common to EIE,ECE,ETE & EEE)			

Q.No		M	BT	CO
1.a	Differentiate between **RISC** and **CISC** architectures and give suitable example for each.	05	2	1
1.b	What is the significance of floating point arithmetic? And represent **$(1163.125)_{10}$** in single and double precision formats	05	3	2
2.a	Describe the evolution of ARM processor families with enhancement of architecture using appropriate timing diagram.	06	2	1
2.b	What is the bottleneck of Von-Neumann Architecture? Illustrate how is it overcome in Harvard architecture? And also compare their performances.	04	2	2
3.a	Is ARM a Processor or Microcontroller? What is ARM **IP licensing**? Differentiate between Microprocessor and Microcontroller with relevant block diagrams.	05	2	2
3.b	After executing the below snippet code, sketch the contents of registers and flags. **i)** **mov r0, #0x47** **mov r1, #0x25** **subs ro, r1** **ii)** **mov r0, #0x47** **mov r1, #0x25** **cmp ro, r1**	05	3	3
4.a	Write a complete ARM Cortex-M assembly language program to find the sum of **six bytes** stored in data area called **X**. Store the final sum in **R5** register and carry in **R6** register. Also write an algorithm for the same.	06	4	3
4.b	List any six salient features of ARM Cortex-M processors and mention the three cores of cortex processors.	04	1	1
5.a	Write a complete ARM Cortex-M assembly language program to interchange byte data between two blocks of memory with starting address **0x20000000 and 0x20000020** respectively. Assume block length is **6**.	05	4	3
5.b	Mention the types of Instruction set in **ARM ISA** and discuss their performances.	05	2	2

BT- Bloom's Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	15	19	16	--	04	25	10	11	--	--

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Test - 1 Microcontroller & programming. [EI 243AI]

KBR (Scheme & Solutions) (1)

1 a)

RISC	CISC
→ Reduced Instruction set Computer processor	→ Complex Instruction set Computer processor
→ Fixed Instruction size	→ Variable Instruction size
→ Less addressing modes	→ More addressing modes
→ Many Registers	→ Less Registers
→ Fetch time/Execution time is same for all Instructions	→ Vary with respect to Instructions
→ Ex: 8051 μC	→ 8086 μp

(05)

2 a) Floating point arithmetic: used to represent very large and very small numbers precisely. (01)

$$(1163.125)_{10} = 10010001011.001$$

Normalization: $1.0010001011001 \times 2^{10}$

$S = 0$; $E = 10$; $M = 0010001011001$

Single precision format:

Sign	Exponent	Mantissa (M)	
0	10001001	0010001011001 0	32 bits
1-bit	8-bits	23-bits	

(04)

(05)

Double precision format

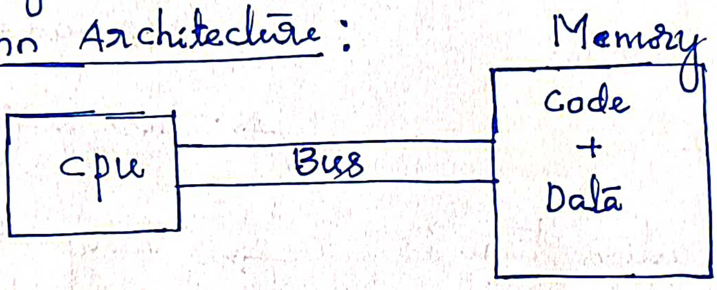
63		0	
0	1000 0001001	0010001011001.........0	64 bits
1-bit	11-bits	52-bits	

2 a) Text book: Fig. 5.1 (06)

2 b) Bottleneck of Von-Neumann Architecture:

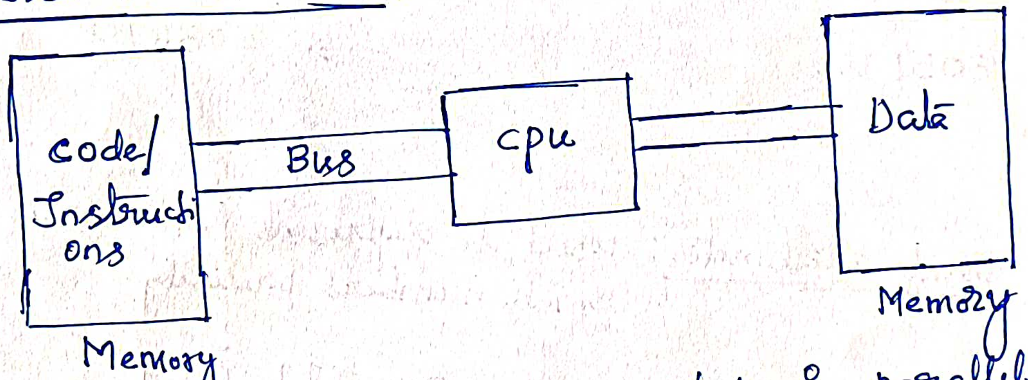
The Instruction/code and data cannot be fetched from memory simultaneously. (01)

Von-Neumann Architecture:



(01)

Harvard Architecture:



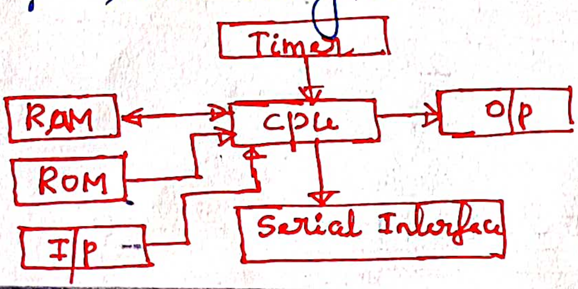
(01)

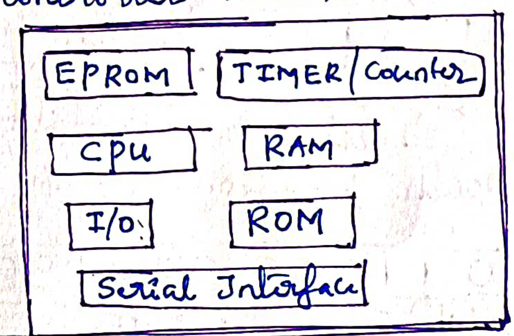
In Harvard architecture, code is executed in parallel with data so it takes less clock cycles. (01)

(04)

3 a) ARM is core for both microprocessors and microcontroller. ARM is a company who license cpu cores. (01)

ARM IP licensing: ARM designs processors and give licenses of processor designs to various microcontroller vendors. (02)





(02)

(05)

µp: cpu and several supporting chips.

µc: System on chip (Soc)

3 b) i) $\boxed{0\times 00000025}$ r_1

$\boxed{0\times 000000de}$ r_0

N=0; Z=0; C=0; V=0

ii) $\boxed{0\times 00000047}$ r_0

$\boxed{0\times 00000025}$ r_1

N=0; Z=0; C=1; V=0

2.5×2 = (05)

4 a) program → 04
Algorithm → 02

04
02
(06)

* program:

```
        export __main
        area prg1, code, readonly

__main
        mov r0, #6
        mov r5, #0
        mov r6, #0
        ldr r1, =x
loop2   adds r5, [r1], #1
        bcc loop1
        add r6, #1
loop1   subs r0, #1
        bne loop2
here    b here
        area x, data, readonly
        dcb 1, 2, 3, 4, 5, 6
        end
```

4 b) **saient features of ARM cortex-M :**

i) Three-stage pipeline design ii) Thumb-2 Technology iii) ARM AMBA Technique iv) Having NVIC v) Having MPU vi) supports Bit banding. (03)

Three cores of cortex processor: (01)

ARM cortex core →
- cortex-A: Application processor core
- cortex-R: Real-time applications core
- cortex-M: Microcontroller core

(04)

5 a)

```
        export __main
        area prg1, code, readonly

__main
        ldr r0, =0x20000000
        ldr r1, =0x20000020
        mov r2, #6
loop1   ldrb r3, [r0]
        ldrb r4, [r1]
        strb r3, [r1], #1
        strb r4, [r0], #1
        subs r2, r2, #1
        bne loop1
here    b here
        end
```

(05)

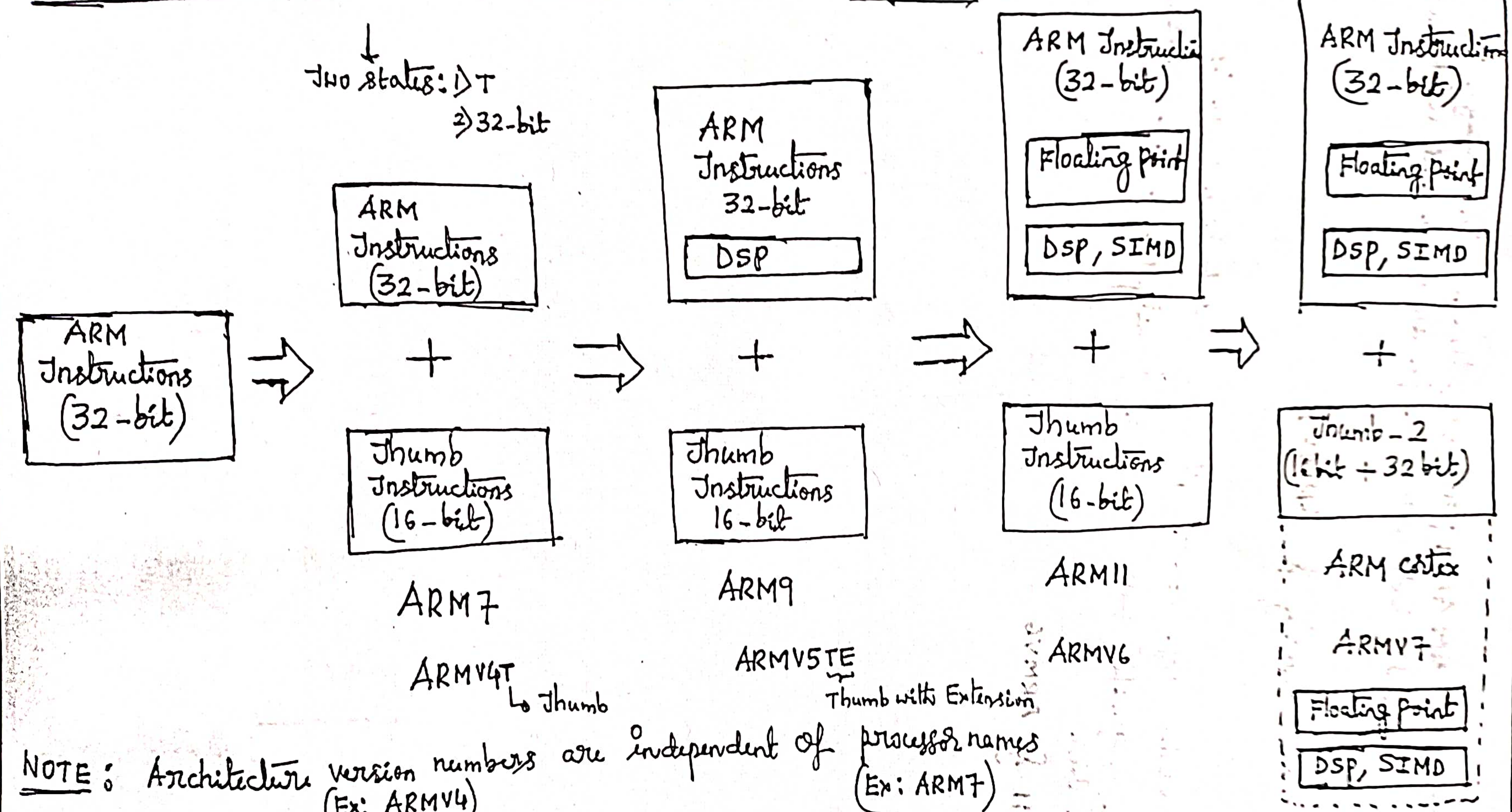
5 b) **Instruction set:** →
- ARM Instruction set: 32-bit long.
- Thumb Instruction set: 16-bit long
- Thumb-2 Instruction set: Mixing of 32-bit and 16-bit Instructions

(05)

Q. 2 a)

Dr. KBR

* Evaluation of ARM Instruction set Architecture (ISA):



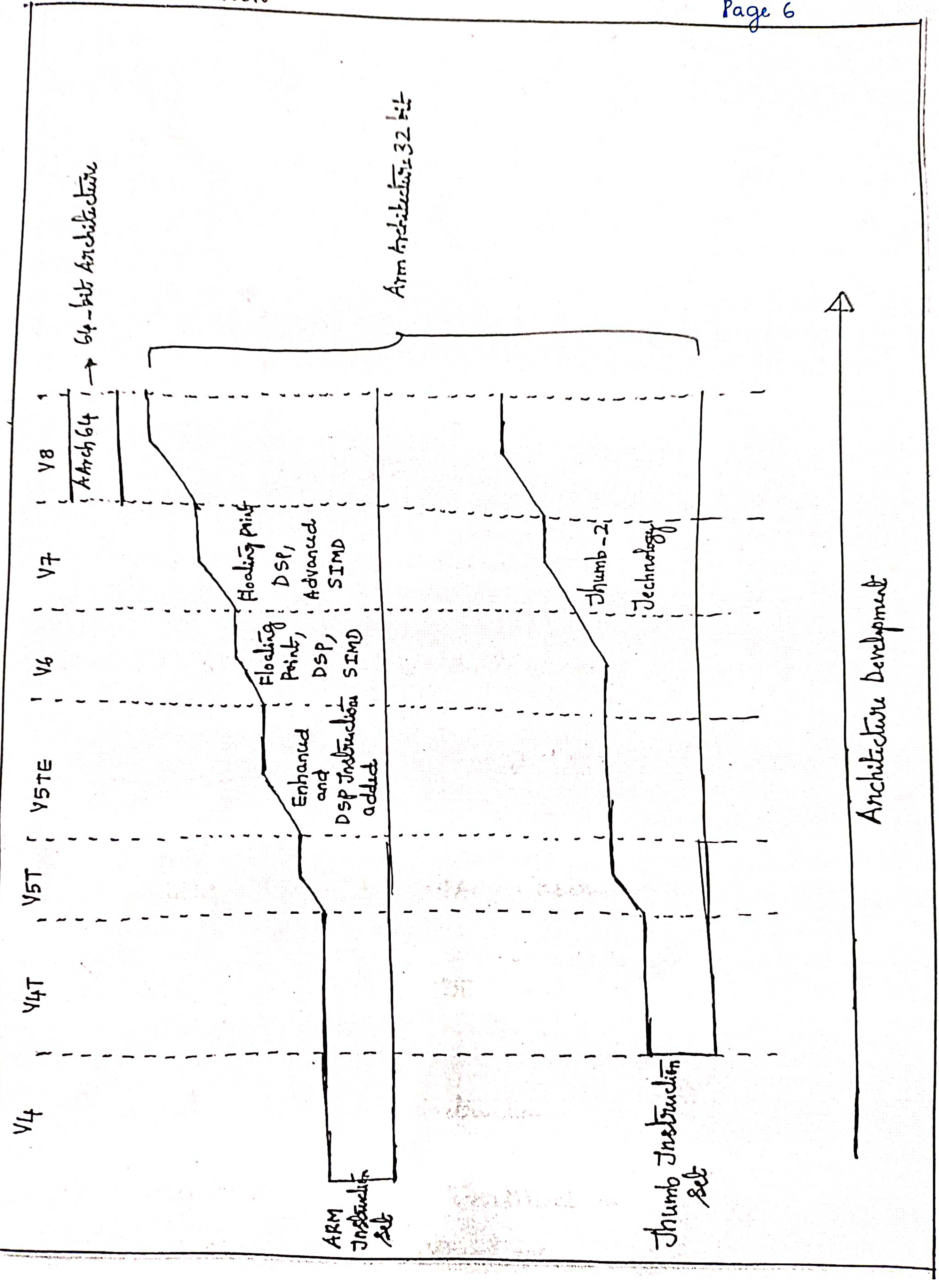
NOTE: Architecture version numbers (Ex: ARMV4) are independent of processor names (Ex: ARM7)

DSP: Digital signal processing operations

SIMD: Single Instruction multiple data

Thumb Instructions was separate in ARM7, ARM9 & ARM11 processors

* In ARM cortex family processors, Thumb Instruction set is merged with ARM Instruction set and there is no separate ARM & Thumb Instruction set, they merged together called it as "Thumb-2" ISA called


V4
V4T
V5T
V5TE
V6
V7
V8
AArch64
64-bit Architecture
Arm Architecture 32 bit
Floating Point, DSP, Advanced SIMD
Floating Point, DSP, SIMD
Enhanced and DSP Instruction added
Thumb-2 Technology
ARM Instruction Set
Thumb Instruction Set
Architecture Development